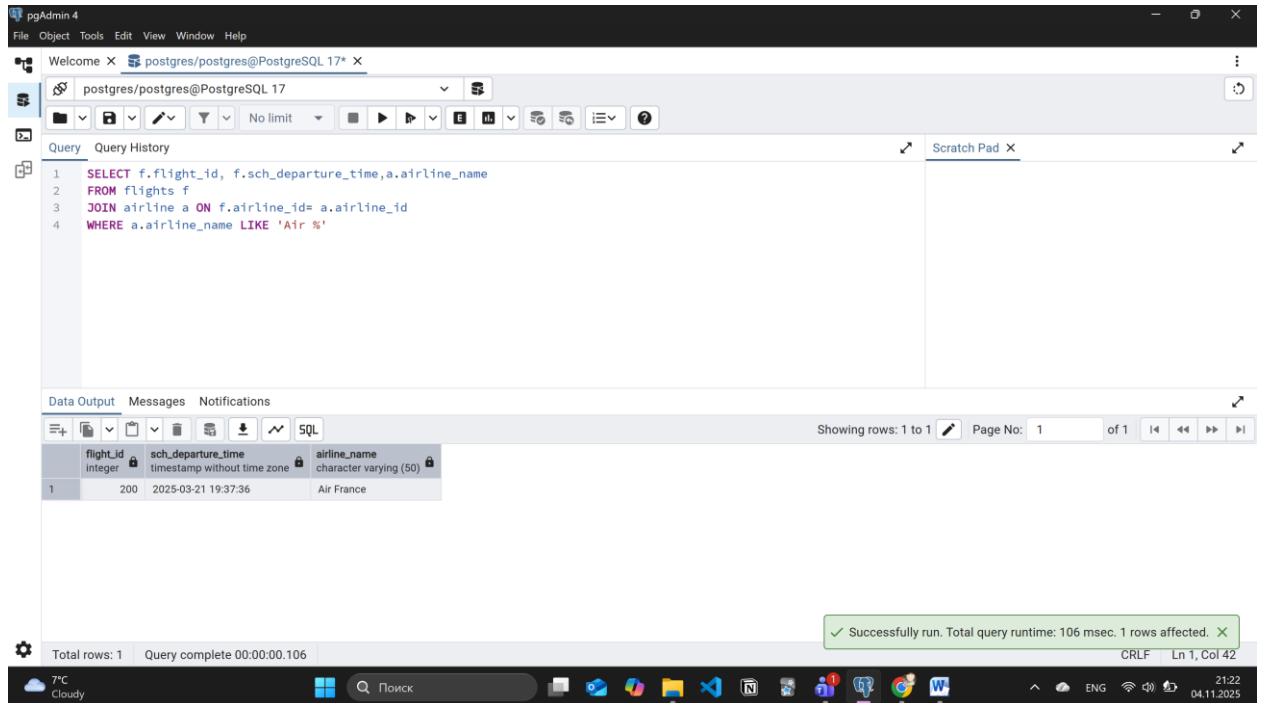


Laboratory work 6

JOIN operations tasks

1. Write a query that displays all flights of a specific airline.



The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is:

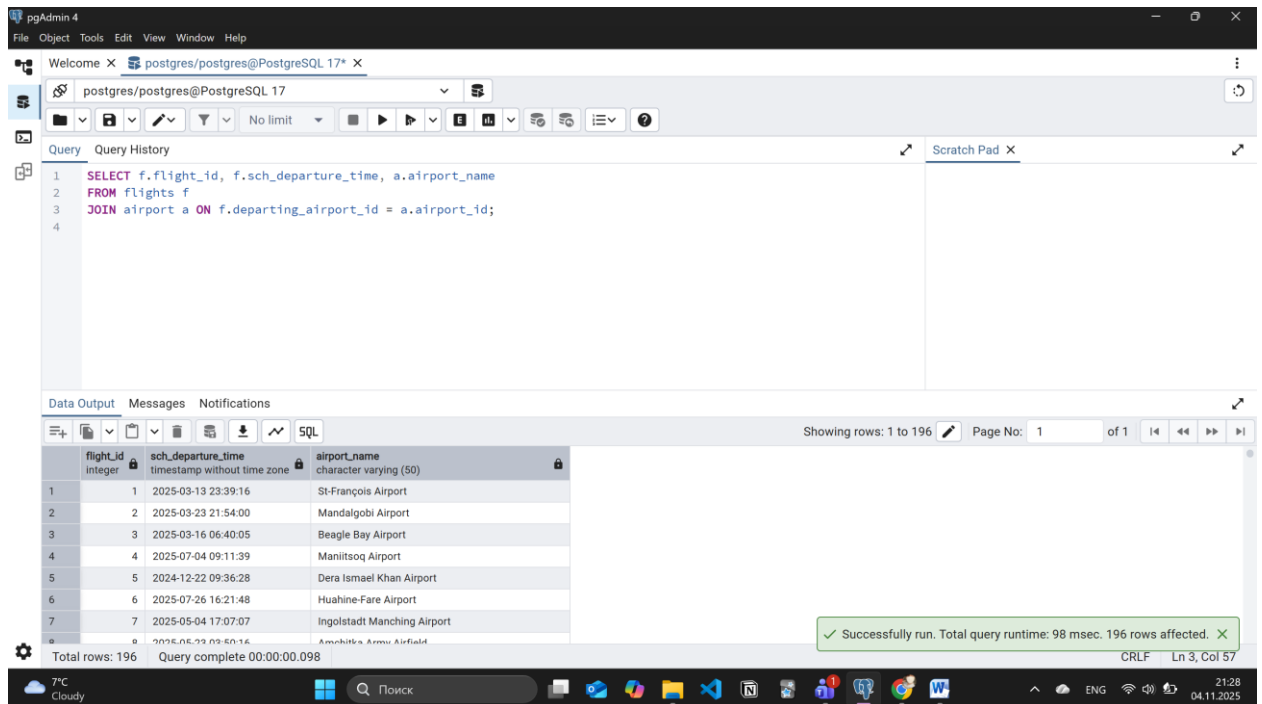
```
1 SELECT f.flight_id, f.sch_departure_time, a.airline_name
2 FROM flights f
3 JOIN airline a ON f.airline_id = a.airline_id
4 WHERE a.airline_name LIKE 'Air %'
```

The results are displayed in a table with the following columns: flight_id (integer), sch_departure_time (timestamp without time zone), and airline_name (character varying (50)).

flight_id	sch_departure_time	airline_name
200	2025-03-21 19:37:36	Air France

The status bar indicates: Total rows: 1, Query complete 00:00:00.106. A message box at the bottom right says: "Successfully run. Total query runtime: 106 msec. 1 rows affected."

2. Compose a query to obtain a list of all flights with the names of departure airports.



The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is:

```
1 SELECT f.flight_id, f.sch_departure_time, a.airport_name
2 FROM flights f
3 JOIN airport a ON f.departing_airport_id = a.airport_id;
```

The results are displayed in a table with the following columns: flight_id (integer), sch_departure_time (timestamp without time zone), and airport_name (character varying (50)).

flight_id	sch_departure_time	airport_name
1	2025-03-13 23:39:16	St-François Airport
2	2025-03-23 21:54:00	Mandalgobi Airport
3	2025-03-16 06:40:05	Beagle Bay Airport
4	2025-07-04 09:11:39	Manitsoq Airport
5	2024-12-22 09:36:28	Dera Ismael Khan Airport
6	2025-07-26 16:21:48	Huahine-Fare Airport
7	2025-05-04 17:07:07	Ingolstadt Manching Airport

The status bar indicates: Total rows: 196, Query complete 00:00:00.098. A message box at the bottom right says: "Successfully run. Total query runtime: 98 msec. 196 rows affected."

3. Create a query that finds all airlines that have no flights scheduled for the next month.

The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is:

```

1 SELECT a.airline_id, a.airline_name
2 FROM airline a
3 LEFT JOIN flights f ON a.airline_id = f.airline_id
4 AND DATE_TRUNC('month', f.sch_departure_time) = DATE_TRUNC('month', CURRENT_DATE + INTERVAL '1 month')
5 WHERE f.flight_id IS NULL;

```

The Data Output tab shows 14 rows of results:

airline_id	airline_name
191	Lufthansa
198	Singapore Airlines
189	Qatar Airways
165	Singapore Airlines
202	AirEasy
195	British Airways
204	FlyFly
107	Lufthansa
107	Lufthansa
107	Lufthansa
107	Lufthansa
107	Lufthansa
107	Lufthansa
107	Lufthansa
107	Lufthansa

A status message at the bottom indicates: "Successfully run. Total query runtime: 149 msec. 14 rows affected."

4. Create a query to display a list of passengers on a specific flight.

The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query is:

```

1 SELECT p.passenger_id, p.first_name, p.last_name, f.flight_id
2 FROM passengers p
3 JOIN booking b ON p.passenger_id = b.passenger_id
4 JOIN flights f ON b.flight_id = f.flight_id
5 WHERE
6     f.flight_id = 101;

```

The Data Output tab shows the column headers for the result set:

passenger_id	first_name	last_name	flight_id
--------------	------------	-----------	-----------

The status bar at the bottom shows: "Total rows: 0 Query complete 00:00:00.101". A status message at the bottom indicates: "Successfully run. Total query runtime: 101 msec. 0 rows affected."

5. Write a query that calculates the average, total, maximum and minimum price of tickets for each flight.

pgAdmin 4

Welcome X postgres/postgres@PostgreSQL 17* X

postgres/postgres@PostgreSQL 17

Query Query History

```

1 SELECT f.flight_id, AVG(b.ticket_price), SUM(b.ticket_price), MAX(b.ticket_price), MIN(b.ticket_price)
2 FROM flights f
3 JOIN booking b ON f.flight_id = b.flight_id
4 GROUP BY f.flight_id
5 ORDER BY f.flight_id;

```

Data Output Messages Notifications

Showing rows: 1 to 200 Page No: 1 of 1

flightId [PK] integer	avg numeric	sum numeric	max numeric	min numeric
1	3.9900000000000000	3.99	3.99	3.99
2	18.9900000000000000	18.99	18.99	18.99
3	2.4900000000000000	2.49	2.49	2.49
4	4.7900000000000000	4.79	4.79	4.79
5	3.2900000000000000	3.29	3.29	3.29
6	34.9900000000000000	34.99	34.99	34.99
7	45.9900000000000000	45.99	45.99	45.99

Total rows: 200 Query complete 00:00:00.183

Successfully run. Total query runtime: 183 msec. 200 rows affected.

- Create a query that shows all flights flying to a specific country by combining flights, airports and airline, and using the condition on the country name.

pgAdmin 4

Welcome X postgres/postgres@PostgreSQL 17* X

postgres/postgres@PostgreSQL 17

Query Query History

```

1 SELECT f.flight_id, a.airport_name, a.country, al.airline_name
2 FROM flights f
3 JOIN airport a ON f.arriving_airport_id = a.airport_id
4 JOIN airline al ON f.airline_id = al.airline_id
5 WHERE a.country = 'Canada';

```

Data Output Messages Notifications

Showing rows: 1 to 200 Page No: 1 of 1

flight_id integer	airport_name character varying (50)	country character varying (50)	airline_name character varying (50)
-------------------	-------------------------------------	--------------------------------	-------------------------------------

Total rows: 0 Query complete 00:00:00.082

Successfully run. Total query runtime: 82 msec. 0 rows affected.

- Display a list of minor passengers and their arrival destination.

The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query selects passenger details and flight information for passengers under 18 years old.

```

1 SELECT p.passenger_id, p.first_name, p.last_name
2 FROM passengers p
3 JOIN booking b ON p.passenger_id = b.passenger_id
4 JOIN flights f ON b.flight_id = f.flight_id
5 JOIN airport a ON f.arriving_airport_id = a.airport_id
6 WHERE EXTRACT(YEAR FROM AGE(p.date_of_birth)) < 18;

```

The Data Output tab shows the following columns:

passenger_id	first_name	last_name
[PK] integer	character varying (50)	character varying (50)

At the bottom, a status bar indicates: "Successfully run. Total query runtime: 162 msec. 0 rows affected."

- Display the passenger's full name, passport number, and the passenger's current time of arrival at the destination.

The screenshot shows the pgAdmin 4 interface with a SQL query executed. The query selects passenger details, passport number, and arrival time.

```

1 SELECT p.first_name || ' ' || p.last_name AS full_name, p.passport_number, f.sch_arrival_time
2 FROM passengers p
3 JOIN booking b ON p.passenger_id = b.passenger_id
4 JOIN flights f ON b.flight_id = f.flight_id
5 JOIN airport a ON f.arriving_airport_id = a.airport_id;

```

The Data Output tab shows the following columns:

full_name	passport_number	sch_arrival_time
text	character varying (20)	timestamp without time zone

At the bottom, a status bar indicates: "Successfully run. Total query runtime: 92 msec. 0 rows affected."

- Print a list of flights where the airline's home country and origin country are the same. Group them by the airport country.

pgAdmin 4

File Object Tools Edit View Window Help

Welcome X postgres/postgres@PostgreSQL 17* X

postgres/postgres@PostgreSQL 17

No limit

Query Query History

```
1 SELECT a.country, f.flight_id, al.airline_name
2 FROM flights f
3 JOIN airline al ON f.airline_id = al.airline_id
4 JOIN airport a ON f.departing_airport_id = a.airport_id
5 WHERE al.airline_country = a.country
6 ORDER BY a.country;
```

Scratch Pad X

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	country character varying (50)	flight_id integer	airline_name character varying (50)
1	Great Britain	189	Qatar Airways
2	UAE	197	Lufthansa

Total rows: 2 Query complete 00:00:00.089

Successfully run. Total query runtime: 89 msec. 2 rows affected.

CRLF Ln 6, Col 20

7°C Cloudy

Поиск

ENG 22:03 04.11.2025